

Quiz 8- Solutions

April 8, 2026 7:58 PM

MATH 3042- Quiz 8

Full Name:

Time: 20 minutes

Q1. A manufacturer claims that **at least 95%** of products are defect-free.

In a sample of 150 items:

138 are defect-free

Test the claim at $\alpha = 0.01$.

$qnorm(0.95) = 1.644854$ $qnorm(0.975) = 1.959964$ $qnorm(0.01) = -2.326348$ $qnorm(0.005) = -2.57582$

Please show all steps including stating hypotheses, test statistic, p-value, decision, and conclusion. Draw the p-value.

$$H_0: p = 95\% \quad \text{left-tailed test}$$

$$H_A: p < 95\%$$

$$n = 150$$

$$\hat{p} = \frac{138}{150}$$

$$np = 142.5 > 10 \checkmark$$

$$nq = 7.5 < 0 \quad \ddot{}$$

Note: the normal approximation may or may not be producing reliable results here

$$\text{test stat } z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.92 - 0.95}{\sqrt{\frac{0.95 \times 0.05}{150}}} = -1.6858$$

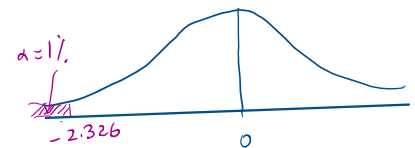
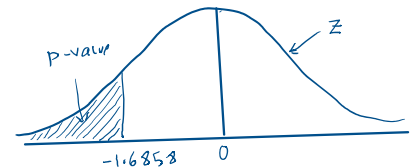
$$p\text{-value} = P(Z \leq -1.6858) = pnorm(-1.6858) = \text{don't know}$$

p-value is greater than α .

we fail to reject H_0

The company's claim of having at least 95% defect-free products is not rejected.

Their true defect rate is not significantly below 95%.



$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} \quad z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

Q2. A sample of waiting times (minutes):

8, 7, 9, 9, 10, 11, 8, 10

$$\bar{x} = 9$$

Test at $\alpha = 0.05$ whether the mean waiting time is different from **10 minutes**.

RStudio output:

```
> wait_time = c(8, 7, 9, 9, 10, 11, 8, 10)
> t.test(wait_time, mu = 10, alternative = "less", conf.level = 0.90 )
```

One Sample t-test

```
data: wait_time
t = -2.1602, df = 7, p-value = 0.03379
alternative hypothesis: true mean is less than 10
90 percent confidence interval:
 -Inf 9.654983
sample estimates:
mean of x
9
```

$$H_0: \mu = 10 \text{ minutes}$$

$$H_A: \mu \neq 10 \text{ minutes}$$

Two sided test

Please show all steps including stating hypotheses, test statistic, p-value, decision, and conclusion. Draw your p-value.

test statistic: -2.1602 (same as one-sided test)

$$P\text{-value: } 2(0.03379) = 0.06758$$

Decision: The p-value is not less than α
No fail to reject H_0

Conclusion: At a 5% significance level, the true mean waiting time is not significantly different from 10 minutes

